Bundle 4 — Parameter search comparison and mechanism-testing tools

Scrum-master view

Bundle objective: prepare the search and comparison layer for the NCA-ART-GRN research engine.

Bundle 3 made the core mechanism loop possible:

```
DSL candidate
  -> PDE/ODE simulation
  -> NCA local rule
  -> ART2 prototypes
  -> ARTMAP transitions
  -> prototype-to-DSL mapping
  -> mechanism report
```

Bundle 4 makes that loop **searchable, comparable, and scientifically testable**.

The goal is not just:

```
find a candidate that makes a nice final pattern
```

The correct goal is:

```
compare search methods by how well they find candidate mechanisms that:
  produce plausible dynamics,
  survive perturbation checks,
  generate useful ART2/ARTMAP structure,
  map back into DSL,
  and suggest discriminating experimental tests.
```

This follows the Hiscock/Megason constraint: final periodic pattern similarity is not enough, because different molecular, cellular, and mechanical mechanisms can produce similar final patterns. Search must therefore optimize for mechanism evidence, dynamics, perturbation response, and falsifiable predictions, not only visual pattern score.

Where this bundle sits in the platform

Bundle 4 is still mostly **config-managed research-engine preparation**.

```
config
  prepares search templates, configs, result schemas, folders, smoke commands

nca-art-grn repo
  owns search code and comparison logic

/workspace/runs/nca-art-grn
  stores search runs

/workspace/artifacts/nca-art-grn
  stores ranked candidates, comparison reports, figures, summaries

Agentfield later
  orchestrates search campaigns as experiments

Paperclip later
  shows campaign status, candidate rankings, reports, and review decisions
```

So these are **not Agentfield steps yet**.
They are `config` bootstrap/check/smoke steps that prepare the repo for future Agentfield orchestration.

User story

```
As the Research Scientist,
I want prepared parameter-search and comparison tools,
so that I can compare random/grid, Latin hypercube, evolutionary, Bayesian,
and robustness-driven searches for 5-node GRN mechanisms, using not only final
```

```
pattern scores but also dynamics, NCA agreement, ART2/ARTMAP evidence,
DSL recoverability, perturbation response, and experimental-design value.
```

Concretizations / managed steps

```
install_parameter_search_comparison_stack

prepare_search_parameter_space
prepare_random_grid_baselines
prepare_lhs_search_template
prepare_evolutionary_search_template
prepare_bayesian_search_template
prepare_active_learning_search_template

prepare_mechanism_scoring_runtime
prepare_search_result_comparison_schema
prepare_candidate_ranking_runtime
prepare_robustness_sweep_template
prepare_perturbation_search_template
prepare_search_report_runtime

prepare_search_smoke_configs
check_parameter_search_inputs
run_parameter_search_local_smoke
```

Step type legend

```
CONFIG STEP
  Managed bootstrap/check/smoke step invoked by config.
```

```
REPO CODE
  Python modules, configs, tests, scripts inside nca-art-grn.

RESEARCH OUTPUT
  Data, run outputs, ranked candidates, metrics, reports.

AGENTFIELD LATER
  Future orchestration layer; not implemented in this bundle.
```

What each step should do

`install_parameter_search_comparison_stack`

Type: CONFIG STEP
Owner: `config` installs packages into the `researchscientist` Python environment.
Target role: `researchscientist` .

Expected package areas:

```
optuna
scikit-optimize
deap
SALib
scipy
scikit-learn
pandas
numpy
networkx
pyyaml
pydantic
rich
tqdm
```

Optional later:

```
nevergrad
botorch
ray
dask
```

What this does for the research:
Gives the research role the tools to compare different ways of searching 5-node GRN parameter/topology space. The point is not only speed. The point is to know which search method finds mechanisms that are plausible, robust, interpretable, and experimentally testable.

How it links into the platform:

```
config installs package stack
repo search modules use those packages
search outputs go to /workspace/runs/nca-art-grn
Agentfield later can run the same search methods as campaign modes
```

`prepare_search_parameter_space`

Type: CONFIG STEP
Owner: `config` prepares schema/config placeholders; repo owns implementation.
Creates/validates REPO CODE:

```
src/nca_art_grn/search/parameter_space.py
src/nca_art_grn/search/constraints.py
src/nca_art_grn/search/sampling_units.py
configs/search/parameter_space_5node.yaml
configs/search/search_constraints.yaml
```

The parameter space must encode:

```
5-node GRN topology variables
activation/repression signs
edge presence/absence
interaction matrix ranges
reaction parameter ranges
diffusion parameter ranges
initial condition families
boundary condition families
perturbation parameters
known 2/3/4-node motif seeds
motif embedding rules
biological plausibility constraints
candidate budget
simulation budget
NCA training/evaluation budget
ART2 vigilance ranges
ARTMAP transition settings
```

What this does for the research:
This defines what is searchable. It keeps the search from becoming random numeric noise by tying parameters to DSL topology, motif priors, biological constraints, simulator settings, NCA settings, ART2 settings, and mechanism-testing needs.

How it links into the platform:

```
DSL candidate runtime defines candidate representation
search_parameter_space defines allowed variations
search methods sample from this space
runs write sampled candidates and parameter metadata
Agentfield later passes this search config into campaign experiments
```

`prepare_random_grid_baselines`

Type: CONFIG STEP
Creates/validates REPO CODE:

```
src/nca_art_grn/search/random_grid.py
configs/search/random_grid_baseline.yaml
```

The baseline must support:

```
fixed grid over small parameter subsets
random candidate sampling
random seed control
candidate budget
simulation budget
repeat count
```

What this does for the research:
Gives you the dumb but honest baseline. Every smarter method must beat this in finding candidates that produce meaningful dynamics, not just final images.

How it links into the platform:

```
random/grid produces baseline candidate set
Bundle 3 pipeline evaluates candidates
comparison report uses this as lower-bound reference
Agentfield later can run baseline campaigns for reproducibility
```

`prepare_lhs_search_template`

Type: CONFIG STEP
Creates/validates REPO CODE:

```
src/nca_art_grn/search/lhs.py
configs/search/latin_hypercube.yaml
```

The Latin hypercube template must encode:

```
parameter ranges
number of samples
stratification seed
continuous parameter coverage
optional topology fixed / topology varied mode
```

What this does for the research:
LHS gives better coverage of continuous GRN parameter space than naive random sampling. It is useful early when you do not yet know which parameter dimensions matter.

How it links into the platform:

```
LHS samples candidates
simulator/NCA/ART2/ARTMAP evaluate them
metrics compare coverage versus discovery quality
Bundle 5 later can scale LHS batches on Runpod
```

`prepare_evolutionary_search_template`

Type: CONFIG STEP
Creates/validates REPO CODE:

```
src/nca_art_grn/search/evolutionary.py
src/nca_art_grn/search/mutation.py
src/nca_art_grn/search/crossover.py
```

The evolutionary search must encode:

```
candidate genome representation
mutation of edge signs
mutation of edge strengths
mutation of diffusion parameters
mutation of reaction parameters
motif-preserving mutation
motif-breaking mutation, optional
selection score
population size
generation count
elitism policy
diversity preservation
```

What this does for the research:
This is where candidate GRNs can evolve toward better mechanism evidence. It should not only select “pretty patterns”; it should select candidates with useful dynamics, perturbation predictions, ART2/ARTMAP structure, and DSL recoverability.

How it links into the platform:

```
candidate DSL -> genome
evolutionary search mutates genome
Bundle 3 evaluates offspring
ranked candidates go to artifacts
Agentfield later can run generations as managed campaign steps
```

`prepare_bayesian_search_template`

Type: CONFIG STEP
Creates/validates REPO CODE:

```
src/nca_art_grn/search/bayesian.py
src/nca_art_grn/search/surrogate_objectives.py
configs/search/bayesian.yaml
```

Bayesian search must encode:

```
objective metric set
continuous parameter bounds
categorical encoding for topology/signs
initial random/LHS points
acquisition function
candidate batch size
budget limit
failed-run handling
```

What this does for the research:
Bayesian optimization helps when simulations/NCA/ART evaluations are expensive. It learns where promising candidates may exist based on previous runs.

How it links into the platform:

```
previous run metrics -> Bayesian surrogate
surrogate proposes next candidates
Bundle 3 evaluates candidates
Bundle 5 later scales expensive batches
Agentfield later can checkpoint campaign state
```

`prepare_active_learning_search_template`

Type: CONFIG STEP
Creates/validates REPO CODE:

```
src/nca_art_grn/search/active_sampling.py
src/nca_art_grn/search/uncertainty.py
configs/search/active_learning.yaml
```

Active learning must encode:

```
uncertainty over mechanism class
uncertainty over perturbation response
uncertainty over prototype assignment
uncertainty over DSL mapping
candidate diversity objective
next-experiment suggestion
```

What this does for the research:
This makes search more scientific: it asks which candidate or perturbation would teach the most about the mechanism, not only which candidate currently scores highest.

How it links into the platform:

```
mechanism reports -> uncertainty signals
active sampler -> next candidate or perturbation
Agentfield later can schedule "most informative next run"
Paperclip later can present suggested next experiments for approval
```

`prepare_mechanism_scoring_runtime`

Type: CONFIG STEP
Creates/validates REPO CODE:

```
src/nca_art_grn/search/scoring.py
src/nca_art_grn/search/objectives.py
```

```
src/nca_art_grn/search/mechanism_score.py
```

The scoring runtime must include multiple score families:

```
final pattern score
pattern dynamics score
wavelength stability score
mode growth score
NCA agreement score
ART2 prototype quality score
ARTMAP transition consistency score
prototype-to-DSL recoverability score
perturbation response score
robustness score
mechanism-discrimination value
experimental-design usefulness
```

What this does for the research:
This is one of the most important Bundle 4 steps. It encodes the updated scientific principle: final pattern is one weak signal, not the final criterion. The search score must prefer candidates that help test mechanisms.

How it links into the platform:

```
Bundle 3 outputs metrics/artifacts
mechanism_scoring_runtime combines them
search methods optimize this score
comparison reports explain score components
Agentfield later exposes score fields in experiment status
```

`prepare_search_result_comparison_schema`

Type: CONFIG STEP
Creates/validates REPO CODE + OUTPUT CONTRACT:

```
src/nca_art_grn/search/result_schema.py
configs/search/result_schema.yaml
```

The result schema must record:

```
search_method
candidate_id
mechanism_hypothesis_id
seed
parameter_set
topology_summary
motif_provenance
simulation_status
NCA_status
ART2_status
ARTMAP_status
pattern_metrics
dynamics_metrics
perturbation_metrics
mechanism_score
failure_reason
artifact_paths
report_path
```

What this does for the research:
Makes different search strategies comparable. Without a shared schema, you cannot fairly compare random search, LHS, evolutionary search, Bayesian optimization, and active learning.

How it links into the platform:

```
all search methods write same result schema
comparison report reads same schema
Agentfield later uses schema as run/campaign status payload
Paperclip later displays method comparison and candidate ranking
```

`prepare_candidate_ranking_runtime`

Type: CONFIG STEP
Creates/validates REPO CODE:

```
src/nca_art_grn/search/ranking.py
src/nca_art_grn/search/pareto.py
configs/search/ranking.yaml
```

Ranking must support:

```
single composite score
multi-objective ranking
Pareto front
minimum viability filters
diversity-aware ranking
mechanism-class grouping
robustness-first ranking
experimental-design-first ranking
```

What this does for the research:
Prevents one metric from dominating. For example, a candidate with slightly weaker final pattern but excellent perturbation-discrimination value may be more scientifically useful than a prettier candidate.

How it links into the platform:

```
search result schema -> ranking
ranking -> candidate shortlist
shortlist -> mechanism reports
shortlist -> Bundle 10 paper figures/tables later
Agentfield later can decide which candidates advance to next campaign
```

`prepare_robustness_sweep_template`

Type: CONFIG STEP
Creates/validates REPO CODE/CONFIGS:

```
src/nca_art_grn/search/robustness.py
configs/search/robustness_sweep.yaml
```

Robustness sweep dimensions:

```
initial condition noise
parameter jitter
diffusion parameter changes
reaction parameter changes
boundary condition changes
grid size changes
time-step changes
perturbation recovery
NCA rollout length
ART2 vigilance variation
ARTMAP transition threshold variation
```

What this does for the research:
Tests whether candidates are stable mechanisms or fragile accidents. Robustness matters because the future paper should argue about mechanisms, not one lucky simulation.

How it links into the platform:

```
candidate shortlist -> robustness sweep
robustness outputs -> mechanism report
Bundle 5 later scales robustness sweeps on Runpod
Agentfield later orchestrates robustness campaigns
```

`prepare_perturbation_search_template`

Type: CONFIG STEP

Creates/validates REPO CODE/CONFIGS:

```
src/nca_art_grn/search/perturbation_search.py
configs/search/perturbation_search.yaml
```

Perturbation search must support:

```
diffusion scaling perturbations
degradation / half-life perturbations
reaction sensitivity perturbations
boundary condition perturbations
initial-condition bias
local ablation / local state reset
noise or heterogeneity injection
NCA-rule perturbation
ART2-vigilance perturbation
```

What this does for the research:

This directly operationalizes Hiscock/Megason: the model should suggest perturbations that distinguish mechanisms. The search should therefore find not just candidates, but candidates with informative perturbation predictions.

How it links into the platform:

```
mechanism hypothesis -> perturbation search
perturbation search -> test configs
Bundle 3 evaluates perturbations
mechanism report records falsification tests
Agentfield later can schedule perturbation campaigns
```

`prepare_search_report_runtime`

Type: CONFIG STEP

Creates/validates REPO CODE + OUTPUT PATHS:

```
src/nca_art_grn/search/report.py
src/nca_art_grn/viz/search_reports.py
/workspace/artifacts/nca-art-grn/search_reports/
```

Reports should include:

```
search method comparison
candidate count
failure count
best candidates
Pareto front
score component breakdown
robustness summary
perturbation summary
NCA agreement summary
ART2/ARTMAP evidence summary
prototype-to-DSL success/failure
recommended next experiments
```

What this does for the research:

Turns search results into scientific decision material. It helps decide which candidate is worth deeper simulation, Runpod scaling, experimental-design writing, or paper inclusion.

How it links into the platform:

```
search outputs -> search report
search report -> Bundle 9 alloy notes
search report -> Bundle 10 paper figures/tables
```

```
Agentfield later attaches report path to campaign status
```

`prepare_search_smoke_configs`

Type: CONFIG STEP
Creates REPO CONFIGS:

```
configs/search/smoke_random_grid.yaml
configs/search/smoke_lhs.yaml
configs/search/smoke_evolutionary.yaml
configs/search/smoke_bayesian.yaml
configs/search/smoke_comparison.yaml
```

Smoke settings must be tiny:

```
1-3 candidates
8x8 or 16x16 grid
few time steps
no Runpod
no full NCA training
tiny ART2/ARTMAP pass
single perturbation check
write smoke comparison report
```

What this does for the research:
Confirms the search wiring works before spending time or money.

How it links into the platform:

```
config invokes tiny search smoke
repo writes comparison output
Agentfield later reuses same config style for real campaigns
```

`check_parameter_search_inputs`

Type: CONFIG CHECK STEP
Runs: no simulations, no training, no search.

Should report:

```
repo exists
research Python env ready
search packages importable
DSL runtime present
mechanism hypothesis configs present
simulator configs present
NCA configs present
ART2/ARTMAP configs present
parameter space config present
scoring config present
result schema present
search output folders present
```

What this does for the research:
Prevents running broken campaigns and makes missing parts visible.

How it links into the platform:

```
operator checks readiness
researchscientist fixes missing inputs
Agentfield later can run equivalent preflight before campaign scheduling
```

`run_parameter_search_local_smoke`

Type: CONFIG SMOKE EXECUTION STEP
Runs: tiny local search only.

Should run:

```
load tiny search config
sample 1-3 candidates
run tiny Bundle 3 evaluation for each
calculate score components
write shared result schema
rank candidates
write smoke comparison report
```

Output:

```
/workspace/runs/nca-art-grn/search_smoke/<timestamp>/
  search_config.yaml
  candidates/
  results.jsonl
  ranking.json
  search_report.md
```

What this does for the research:
Proves that search methods can call the NCA-ART-DSL mechanism loop and produce comparable outputs.

How it links into the platform:

```
today:
  config runs tiny local smoke

later:
  Agentfield runs real search campaign
```

```
later:
```

Whole-system linkage

Bundle 4 should connect like this:

```
Bundle 3
  evaluates one candidate mechanism deeply

Bundle 4
  chooses and compares many candidates/search methods

Bundle 5
  scales expensive searches/training/inference on Runpod

Agentfield later
  orchestrates search campaigns

Paperclip later
  makes campaigns human-reviewable
```

Concrete later flow:

```
config:
  sudo config --target researchscientist bootstrap step run_parameter_search_local_smoke

repo:
  python -m nca_art_grn.cli.compare_search \
    --config configs/search/smoke_comparison.yaml

Agentfield later:
  GRNExperiment.spec.method = parameter_search_comparison
  GRNExperiment.spec.search_method = lhs | evolutionary | bayesian
```

```
  controller calls repo CLI
  status points to /workspace/runs/nca-art-grn/<campaign_id>

Paperclip later:
  shows candidates, method comparison, score breakdown, reports, next-experiment suggestions
```

Acceptance criteria

The following should be valid:

```
sudo config --target researchscientist bootstrap step install_parameter_search_comparison_stack
sudo config --target researchscientist bootstrap step prepare_search_parameter_space
sudo config --target researchscientist bootstrap step prepare_random_grid_baselines
sudo config --target researchscientist bootstrap step prepare_lhs_search_template
sudo config --target researchscientist bootstrap step prepare_evolutionary_search_template
sudo config --target researchscientist bootstrap step prepare_bayesian_search_template
sudo config --target researchscientist bootstrap step prepare_active_learning_search_template
sudo config --target researchscientist bootstrap step prepare_mechanism_scoring_runtime
sudo config --target researchscientist bootstrap step prepare_search_result_comparison_schema
sudo config --target researchscientist bootstrap step prepare_candidate_ranking_runtime
sudo config --target researchscientist bootstrap step prepare_robustness_sweep_template
sudo config --target researchscientist bootstrap step prepare_perturbation_search_template
sudo config --target researchscientist bootstrap step prepare_search_report_runtime
sudo config --target researchscientist bootstrap step prepare_search_smoke_configs
sudo config --target researchscientist bootstrap step check_parameter_search_inputs
```

Explicit tiny execution:

```
sudo config --target researchscientist bootstrap step run_parameter_search_local_smoke
```

Prepare/check steps must:

```
not run full searches
not run large simulations
not train full NCA models
not launch Runpod
not overwrite research code
not overwrite existing search results
not claim final mechanism from final pattern score alone
not build paper outputs
```

Smoke step must:

```
use tiny local configs
run only 1-3 candidates
record search method
record candidate DSL
record mechanism hypothesis
record score components
record failure reasons
write result schema
write search report
clearly label outputs as smoke
```

Proposed tests

Registry and shell syntax tests

```
bash --noprofile --norc -n /home/vmuser/.local/bin/config.sh
bash --noprofile --norc -n /home/vmuser/.local/lib/config-sh/installers.sh
```

```
config bootstrap steps | grep prepare_search_parameter_space
config bootstrap steps | grep prepare_lhs_search_template
config bootstrap steps | grep prepare_evolutionary_search_template
config bootstrap steps | grep prepare_bayesian_search_template
config bootstrap steps | grep prepare_mechanism_scoring_runtime
config bootstrap steps | grep prepare_search_result_comparison_schema
config bootstrap steps | grep run_parameter_search_local_smoke
```

Repo structure tests

```
test -d /workspace/repos/nca-art-grn/src/nca_art_grn/search
test -d /workspace/repos/nca-art-grn/configs/search
test -d /workspace/artifacts/nca-art-grn/search_reports
```

Config tests

```
sudo config --target researchscientist bootstrap step prepare_search_smoke_configs

test -f /workspace/repos/nca-art-grn/configs/search/smoke_random_grid.yaml
test -f /workspace/repos/nca-art-grn/configs/search/smoke_lhs.yaml
test -f /workspace/repos/nca-art-grn/configs/search/smoke_evolutionary.yaml
test -f /workspace/repos/nca-art-grn/configs/search/smoke_bayesian.yaml
test -f /workspace/repos/nca-art-grn/configs/search/smoke_comparison.yaml
```

Scoring schema tests

```
sudo config --target researchscientist bootstrap step prepare_mechanism_scoring_runtime
sudo config --target researchscientist bootstrap step prepare_search_result_comparison_schema

test -f /workspace/repos/nca-art-grn/configs/search/scoring.yaml
test -f /workspace/repos/nca-art-grn/configs/search/result_schema.yaml
```

The scoring config should include fields for:

```
final_pattern_score
pattern_dynamics_score
nca_agreement_score
art2_prototype_quality_score
artmap_transition_consistency_score
perturbation_response_score
mechanism_discrimination_value
```

Readiness check

```
sudo config --target researchscientist bootstrap step check_parameter_search_inputs
```

Expected: readiness report only, no simulations.

Local smoke test

```
sudo config --target researchscientist bootstrap step run_parameter_search_local_smoke

find /workspace/runs/nca-art-grn/search_smoke -name results.jsonl | tail -n 1
find /workspace/runs/nca-art-grn/search_smoke -name ranking.json | tail -n 1
find /workspace/runs/nca-art-grn/search_smoke -name search_report.md | tail -n 1
```

Scientific guardrail test

The smoke search report must contain:

```
Search method
Candidate IDs
Score component breakdown
Final pattern score is not sufficient
```

```
Dynamics evidence
Perturbation evidence
Mechanism-discrimination value
Recommended next experiment
```

Non-overwrite test

```
echo "DO NOT OVERWRITE" > /workspace/repos/nca-art-grn/configs/search/scoring.yaml
sudo config --target researchscientist bootstrap step prepare_mechanism_scoring_runtime
grep "DO NOT OVERWRITE" /workspace/repos/nca-art-grn/configs/search/scoring.yaml
```

Summary

Bundle 4 is the **search and comparison layer**.

It does not replace Bundle 3. It repeatedly calls Bundle 3's mechanism-evaluation loop.

```
Bundle 3 asks:
  Is this candidate mechanism meaningful?

Bundle 4 asks:
  Which search method finds meaningful mechanisms efficiently and robustly?
```

Scientifically, Bundle 4 must search for:

```
not just final patterns,
but mechanisms with useful dynamics,
NCA agreement,
ART2 prototype structure,
ARTMAP transition consistency,
DSL recoverability,
```

```
robustness,
perturbation response,
and experimental-design value.
```

Platform-wise:

```
config prepares the search tools
nca-art-grn executes the search logic
/workspace stores runs and reports
Agentfield later orchestrates campaigns
Paperclip later exposes those campaigns for human review
```